An Updated Dataset of Survey-Based Fiscal News Proxies

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Abstract

Proxies for US government spending shocks built from the Survey of Professional Forecasters are widely used to address fiscal foresight, but the series circulated with the original papers end in the early 2010s. This note provides an updated and reproducible version of three families of proxies: the one-quarter-ahead forecast error of Ramey (2011), the expectation and news measures of Forni and Gambetti (2016), and the individual-data news and nowcast-error measures of Ricco (2015).

Keywords: fiscal foresight, government spending shocks, fiscal news, Survey of Professional Forecasters, forecast revisions, aggregation bias
JEL codes: C32, C82, E32, E62.

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The updated instruments can be found at https://markeljanblaceri.github.io/data.

1 Introduction

Fiscal policy act with two lags (Leeper et al., 2013). The inside lag runs from the moment a measure is proposed or announced to the moment it is enacted into law. The outside lag runs from enactment to the moment the money is actually spent and recorded in the national accounts. Together they can last many quarters. A defense build-up or a stimulus package is debated, then legislated, and only then paid out over several years.

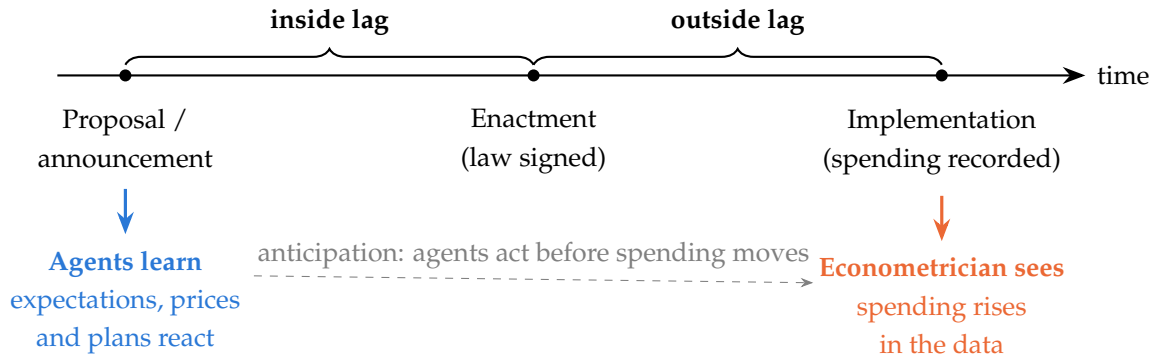


Figure 1: The two lags of fiscal policy

Notes: Agents react when news arrives; the econometrician observes the change only when spending is implemented.

Figure 1 shows why this makes fiscal shocks hard to identify. Households, firms, and markets react when the news arrives, at the left end of the timeline. They adjust consumption, investment, and prices in anticipation. A researcher who looks only at realized spending sees the change at the right end, when the money is spent. By then much of the economic response has already happened, and the spending increase is no longer a surprise to anyone in the economy. A standard structural VAR (e.g. Blanchard and Perotti, 2002), which treats unexpected movements in realized spending as shocks, therefore dates the shock too late and may label as a "shock" something that agents expected for quarters. Technically, agents have more information than the econometrician, a problem known as fiscal foresight, and the VAR innovations may then be non-fundamental (Leeper et al., 2013).

The remedy is to measure what agents know when the news arrives, by bringing their expectations into the econometric model. Survey forecasts of government spending are a natural source. Since 1981 the Survey of Professional Forecasters (SPF) has collected quarterly forecasts of real federal government consumption expenditures

and gross investment, and several proxies for fiscal shocks have been built from them (Ramey, 2011; Forni and Gambetti, 2016; Ricco, 2015). These proxies are used in expectational VARs, as external instruments in proxy-SVARs, and in local projections, but the series circulated with the original papers end around 2012–2013. This note reviews how the proxies are constructed and provides an updated, reproducible version of all of them through 2026. Section 2 reviews the methods, Section 3 describes the data, the construction of the proxies and the updated series, and Section 4 concludes.

2 Identifying fiscal shocks with survey forecasts

Throughout, Δg_t denotes the growth rate of real federal spending in quarter t and $\mathbb{E}_t^*$ the (not necessarily rational) expectation of professional forecasters formed during quarter t. The survey is conducted in the middle of the quarter, so forecasters know the first release of Δg_{t-1} but not Δg_t. The survey therefore contains a nowcast $\mathbb{E}_t^* \Delta g_t$ and forecasts $\mathbb{E}_t^* \Delta g_{t+h}$ for $h = 1, \ldots, 4$.

Forecast errors as fiscal surprises. Ramey (2011) argues that the timing of fiscal shocks is central to their identification. If spending changes are anticipated, the innovations of a standard VAR are not true shocks, and she shows that they are in fact predicted by professional forecasts of government spending. Besides her narrative military news variable, she proposes the one-quarter-ahead SPF forecast error,

$$FE_t = \Delta g_t - \mathbb{E}_{t-1}^* \Delta g_t, \tag{1}$$

as a measure of the unanticipated component of spending, using the median SPF forecast. Auerbach and Gorodnichenko (2012) similarly use forecast errors from the SPF and the Greenbook to purge VAR spending innovations of their predictable component.

A critique. Perotti (2011) observes that the forecast error in (1) mixes two objects with different timing:

$$\underbrace{\Delta g_t - \mathbb{E}_{t-1}^* \Delta g_t}_{\text{forecast error}} = \underbrace{\left(\Delta g_t - \mathbb{E}_t^* \Delta g_t\right)}_{\text{nowcast error}} + \underbrace{\left(\mathbb{E}_t^* \Delta g_t - \mathbb{E}_{t-1}^* \Delta g_t\right)}_{\text{nowcast revision}}. \tag{2}$$

The nowcast error is information that reaches agents only after quarter t. The nowcast revision is news that reaches them during quarter t. The two components can have

distinct macroeconomic effects, so a shock identified from their sum is a blend whose properties depend on their relative variances. Moreover, by construction, (1) only captures spending changes that were not anticipated one quarter ahead. It is silent on news about spending in future quarters, which is precisely the information that fiscal foresight is about.

Expectations and revisions. Forni and Gambetti (2016) model spending growth as driven by three shocks,

$$\Delta g_t = \phi(L)\varepsilon_t + \theta(L)\eta_t + \delta(L)\xi_t, \tag{3}$$

where ε_t is a news (foresight) shock, η_t a surprise shock, and ξ_t collects non-policy shocks to which spending reacts automatically. News affects spending only with a delay $s > 0$, so $\phi_k = 0$ for $k < s$. Following Blanchard and Perotti (2002), non-policy shocks have no contemporaneous effect ($\delta_0 = 0$), whereas the surprise shock does ($\theta_0 \neq 0$). Forecasters observe ε_t immediately and η_t, ξ_t with a one-quarter delay.

The sum of expectations over the next H quarters is

$$F_t(1,H) = \sum_{h=1}^{H} \mathbb{E}_t^* \Delta g_{t+h} = \tilde{\phi}_0 \varepsilon_t + \sum_{j\geq 1} \tilde{\phi}_j \varepsilon_{t-j} + \sum_{j\geq 1} \tilde{\theta}_j \eta_{t-j} + \sum_{j\geq 1} \tilde{\delta}_j \xi_{t-j}, \quad \tilde{\phi}_0 = \sum_{k=s}^{H} \phi_k, \tag{4}$$

and contains the current news shock plus lags of all shocks. Summing over horizons guards against not knowing the anticipation lag s.[1] Alternatively, the sum of revisions,

$$\begin{aligned} n_t(1,H) &= F_t(1,H) - F_{t-1}(2,H+1) \\ &= \sum_{h=1}^{H} (\mathbb{E}_t^* \Delta g_{t+h} - \mathbb{E}_{t-1}^* \Delta g_{t+h}) = \tilde{\phi}_0 \varepsilon_t + \tilde{\theta}_1 \eta_{t-1} + \tilde{\delta}_1 \xi_{t-1}, \end{aligned} \tag{5}$$

contains the current news shock and only the first lag of the other shocks.[2] In the original paper, either proxy is ordered second, after federal spending, in a recursive VAR with GDP, consumption, the federal surplus, net exports, the 10-year Treasury yield and the real exchange rate: the first shock is the surprise and the second the news shock.

[1] Because $F_t(1,H)$ also contains the lags of all shocks, the news shock is the component of $F_t(1,H)$ orthogonal to past information. In practice $F_t(1,H)$ is included in a VAR and the news shock is identified recursively, conditioning on the past of all variables. Forni and Gambetti (2016) call this "Identification #1".

[2] Here too, the news shock is obtained by conditioning on the past of the VAR variables. Forni and Gambetti (2016) call this "Identification #2".

Imperfect information. Ricco (2015) argues that identification strategies derived from full-information rational expectations can misidentify fiscal shocks when agents face information frictions. In sticky- or noisy-information models, average forecasts respond gradually, and average forecast errors are related to past revisions (Coibion and Gorodnichenko, 2012):

$$x_t - \mathbb{E}^*_{t-h} x_t = \frac{1-\kappa}{\kappa}(\mathbb{E}^*_{t-h} x_t - \mathbb{E}^*_{t-h-1} x_t) + u_{t-h+1} + \ldots + u_t, \tag{6}$$

where $\kappa \in (0, 1]$ measures information rigidity ($\kappa = 1$ under full information). Forecast and nowcast errors are then predictable and combine current and past shocks, so they are poor proxies for structural shocks. Revisions of expectations, conditional on past information, are the informative object. Iterating (2) gives

$$\begin{aligned} \underbrace{\Delta g_t - \mathbb{E}^*_{t-h}\Delta g_t}_{\text{forecast error}} = \underbrace{(\Delta g_t - \mathbb{E}^*_t \Delta g_t)}_{\text{nowcast error}} + \underbrace{(\mathbb{E}^*_t \Delta g_t - \mathbb{E}^*_{t-1}\Delta g_t)}_{\text{nowcast revision}} \\ + \ldots + \underbrace{(\mathbb{E}^*_{t-h+1}\Delta g_t - \mathbb{E}^*_{t-h}\Delta g_t)}_{\text{forecast revision}}, \end{aligned} \tag{7}$$

which motivates three types of innovations, each with its own proxy: *misexpected* changes, not perceived when they occur and proxied by the nowcast error $g_t - \mathbb{E}^*_t g_t$; *unexpected* changes, perceived when they occur and proxied by the nowcast revision $\mathbb{E}^*_t g_t - \mathbb{E}^*_{t-1} g_t$; and *expected* changes, anticipated by agents and proxied by revisions of forecasts for future quarters, $\mathbb{E}^*_t g_{t+h} - \mathbb{E}^*_{t-1} g_{t+h}$ for $h \geq 1$.

Aggregation bias. The measures discussed so far are revisions of aggregate (mean or median) forecasts. The key methodological point of Ricco (2015) is that such revisions compare the central tendency of two different cross-sections of forecasters. Let $\mathcal{S}_t$ be the set of the S_t forecasters responding at t. Then

$$\underbrace{\frac{1}{S_t}\sum_{i\in\mathcal{S}_t} \mathbb{E}^{*i}_t \Delta g_{t+h} - \frac{1}{S_{t-1}}\sum_{j\in\mathcal{S}_{t-1}} \mathbb{E}^{*j}_{t-1}\Delta g_{t+h}}_{\text{revision of the aggregate}} \neq \underbrace{\frac{1}{|\mathcal{S}_t \cap \mathcal{S}_{t-1}|}\sum_{k\in\mathcal{S}_t\cap\mathcal{S}_{t-1}} \left(\mathbb{E}^{*k}_t \Delta g_{t+h} - \mathbb{E}^{*k}_{t-1}\Delta g_{t+h}\right)}_{\text{aggregate of individual revisions}}, \tag{8}$$

where $\mathbb{E}^{*i}_t$ is the expectation of forecaster i. The two sides coincide only with a balanced panel and, for medians, symmetric distributions. The SPF departs from both conditions:

1. *Unbalanced panel.* Respondents enter and exit, so changes in composition add noise to aggregate revisions. This is classical measurement error.

2. *Skewed revisions.* The cross-sectional distribution of revisions is skewed, and its skewness is correlated with the size of the news. The error in the aggregate measure is then correlated with the true news: non-classical measurement error.

Non-classical measurement error of this kind biases the estimated effects of news towards zero, beyond classical attenuation. Ricco (2015) supports the argument with Monte Carlo simulations calibrated to the SPF and with forecasting regressions in which individual-data news is more informative about future spending than its aggregate-data counterpart.

Ricco (2015) therefore computes revisions forecaster by forecaster and aggregates them afterwards:

$$\mathcal{M}_t = \Delta g_t - \frac{1}{N_i} \sum_i \mathbb{E}_t^{*i} \Delta g_t \tag{9}$$

$$\mathcal{N}_t(0) = \frac{1}{N_i} \sum_i \left(\mathbb{E}_t^{*i} \Delta g_t - \mathbb{E}_{t-1}^{*i} \Delta g_t \right) \tag{10}$$

$$\mathcal{N}_t(1,3) = \sum_{h=1}^{3} \frac{1}{N_i} \sum_i \left(\mathbb{E}_t^{*i} \Delta g_{t+h} - \mathbb{E}_{t-1}^{*i} \Delta g_{t+h} \right) \tag{11}$$

where revisions use forecasters responding in two consecutive surveys, and $\mathcal{M}_t$ uses all respondents at t. The equations above use the cross-sectional mean; in his main results Ricco (2015) uses the median instead.[3] The innovations to $\mathcal{N}_t(0)$, $\mathcal{M}_t$ and $\mathcal{N}_t(1,3)$ in a large Bayesian VAR identify unexpected, misexpected and expected fiscal changes, respectively. In this light, Ramey's forecast error (1) combines the misexpected and the unexpected components.

3 Empirical proxies

Survey of Professional Forecasters. The SPF is the oldest quarterly survey of US macroeconomic forecasts. It was started in 1968 by the American Statistical Association and the NBER and has been run by the Federal Reserve Bank of Philadelphia

[3] The median versions replace the cross-sectional mean with the median over forecasters:

$$\mathcal{M}_t^{\text{med}} = \Delta g_t - \operatorname*{med}_i \mathbb{E}_t^{*i} \Delta g_t, \qquad \mathcal{N}_t^{\text{med}}(0) = \operatorname*{med}_i \left(\mathbb{E}_t^{*i} \Delta g_t - \mathbb{E}_{t-1}^{*i} \Delta g_t \right),$$

$$\mathcal{N}_t^{\text{med}}(1,3) = \operatorname*{med}_i \sum_{h=1}^{3} \left(\mathbb{E}_t^{*i} \Delta g_{t+h} - \mathbb{E}_{t-1}^{*i} \Delta g_{t+h} \right),$$

that is, the median of each forecaster's summed revisions.

since 1990Q2. Panelists are mostly private-sector forecasters. Each quarter they report forecasts for the previous quarter, the current quarter, and the next four quarters, plus annual averages.

Questionnaires are sent after the BEA advance release of the national accounts at the end of the first month of the quarter, and responses are due in the middle month. Forecasters therefore know the first estimate of quarter $t-1$ when forecasting quarter t, which underlies the definitions of nowcast, nowcast error and revision.

The Philadelphia Fed releases the forecasts in three files, which map into the ways of constructing the proxies:

- Mean and median level forecasts, computed across respondents horizon by horizon. These are used for the proxies of Ramey (2011) and Forni and Gambetti (2016). Because the median is taken separately at each horizon, the median levels at $t+h$ and $t+h+1$ can come from different forecasters.
- Individual responses, identified by an anonymous forecaster ID, which allow each forecaster to be tracked across surveys. These are used for the proxies of Ricco (2015).

Table 1 summarizes the panel of individual responses. Forecasters enter and exit frequently, so only about three quarters of the respondents in a given survey can be matched with the previous one.

Table 1: The SPF panel of individual responses for real federal spending

	1981–2012	2013–2026	1981–2026	Ricco (2015)
Surveys	126	55	181	–
Distinct forecasters	215	71	240	–
Respondents per survey: mean	31.0	32.4	31.4	29
min–max	9–51	23–41	9–51	–
Matched with previous survey: mean	23.3	27.5	24.6	22
min–max	7–47	20–33	7–47	–

Notes: Surveys 1981Q3–2026Q3. Respondents are forecasters with a valid nowcast after cleaning. Matched forecasters also responded to the previous survey and report forecasts at all horizons, the sample used for $\mathcal{N}_t(1,3)$. The figures for Ricco (2015) refer to his sample, 1981–2012.

Spending variable. Forecasts of real federal government consumption expenditures and gross investment are available from the 1981Q3 survey.[4] The SPF reports forecasts of real levels, whose base year changes several times in the sample. Following the literature, all proxies are built from growth rates computed within a survey, which removes the base-year breaks.

Realized growth comes from the Real-Time Data Set for Macroeconomists (Croushore and Stark, 2001), series RGF. We use the latest vintage (October 2026), that is, revised data, as in Forni and Gambetti (2016) and Ricco (2015). Revised data are consistent over the whole sample, whereas real-time vintages from before 1996 refer to the earlier concept of federal government purchases, which the BEA replaced with consumption expenditures and gross investment that year. Realized spending enters only the forecast error and the nowcast error; the news measures depend on survey data alone.

Growth convention. For a row of level forecasts L (from the mean or median file, or from a single forecaster), implied growth is

$$f_t(h) = 100 \left[\left(\frac{L_{t+h}}{L_{t+h-1}} \right)^4 - 1 \right], \quad h = 0, \ldots, 4, \tag{12}$$

where L_{t-1} is the level shown on the questionnaire. $f_t(h)$ is the empirical counterpart of $\mathbb{E}_t^* \Delta g_{t+h}$, and realized growth Δg_t is computed in the same way from the realized levels. Dates refer to the survey quarter, except for Ramey's forecast error, which is dated by the quarter being forecast.

Proxies from aggregate forecasts. Applying (12) to the mean or median levels gives $f_t^{\text{mean}}(h)$ and $f_t^{\text{median}}(h)$, from which:

- Ramey (2011) forecast error: $FE_t = \Delta g_t - f_{t-1}(1)$;
- Forni and Gambetti (2016): $F_t(1,4) = \sum_{h=1}^{4} f_t(h)$ and $n_t(1,3) = F_t(1,3) - F_{t-1}(2,4)$.

Growth rates implied by median levels are noisier than those implied by mean levels, because consecutive median levels may belong to different forecasters, and many early respondents reported rounded integer levels.

[4] The forecasts are available at https://www.philadelphiafed.org/surveys-and-data/rfedgov.

Proxies from individual forecasts. Applying (12) to each forecaster's own levels gives $f_t^i(h)$. For each survey we match forecasters by ID with the previous survey and compute the revisions $\mathcal{N}_t^i(h) = f_t^i(h) - f_{t-1}^i(h+1)$, for $h = 0, \ldots, 3$. For $\mathcal{N}_t(1,3)$ we keep forecasters with all three horizons, so that the panel is balanced across horizons. The revisions are then aggregated by the mean, as in (10)–(11), or by the median. Following Ricco (2015), the median of $\mathcal{N}_t(1,3)$ is the median of each forecaster's revisions summed over the three horizons. The nowcast error $\mathcal{M}_t$ in (9) uses all respondents at t.

With the mean, the Forni and Gambetti (2016) news measure computed on the matched panel of individual forecasts coincides with $\mathcal{N}_t(1,3)$, since $n_t(1,3) = \sum_{h=1}^{3}[f_t(h) - f_{t-1}(h+1)]$ and the mean of differences equals the difference of means. The two approaches therefore differ only in how forecasts are aggregated: by changes in panel composition and, for medians, by the skewness of revisions, which are exactly the two channels in (8).

Cleaning. A small number of individual responses report a $t-1$ level far from the questionnaire value, most likely on a different base year or concept, which creates spurious nowcast growth. For example, one forecaster in 2011–2012 reports levels about 13% above the cross-section, implying nowcasts of about -50% annualized. We drop individual rows whose $t-1$ level deviates by more than 2% from that quarter's median, which removes 44 rows. Medians are essentially unaffected; means are protected from these outliers. The aggregate files are used as published.

3.1 The updated series

Figures 2–4 plot the updated proxies. The post-2012 period adds informative episodes, notably the COVID-19 relief packages of 2020, which produce clear positive news in both families of news measures and the largest forecast and nowcast errors in the sample (2020Q2); smaller positive news spikes appear in 2018Q2 and 2021Q1. Three features stand out. First, Ramey's forecast error and Ricco's nowcast error are almost indistinguishable whether built from the mean or the median, and both are dominated by high-frequency movements that show little correspondence with identifiable fiscal events. Second, news about future quarters is far more concentrated on a few large episodes, consistent with its interpretation as foresight. Third, the median versions of the individual-data news measures are much flatter than the mean versions after

the early 1990s. Revisions cluster around zero (the share of forecasters leaving their forecasts unchanged rises from about 5% in the 1980s to over 20% after 2012), so the median revision moves only when large, widely shared news shifts the whole distribution, whereas the mean also responds to a few large individual revisions.

4 Conclusion

This note provides an updated dataset of survey-based proxies for US government spending shocks from 1981 to 2026. It covers the one-quarter-ahead forecast error of Ramey (2011) and the expectation and news measures of Forni and Gambetti (2016), both built from the mean and the median SPF forecasts, and the news and nowcast-error measures of Ricco (2015), built from individual responses aggregated by the mean and the median. All proxies follow the constructions of the original papers, with a common growth convention, a single source of realized spending, and a simple cleaning rule for inconsistent individual responses. The extension adds more than a decade of data, including the fiscal response to the COVID-19 pandemic.

A few points deserve attention from users. The series are proxies, not identified shocks: recovering the shocks of the original papers requires re-estimating their VARs. The forecast and nowcast errors depend on the vintage of realized spending and can change slightly when the BEA revises the national accounts, whereas the news measures depend on survey data alone. The 2020 observations are extreme in both forecasts and realizations, so estimation over the full sample may call for dummies, a sample ending in 2019, or volatility adjustments. Finally, the choice between mean and median aggregation is not innocuous, especially for news about future quarters, which is why both are reported. The dataset is available online and will be updated as new SPF releases become available.

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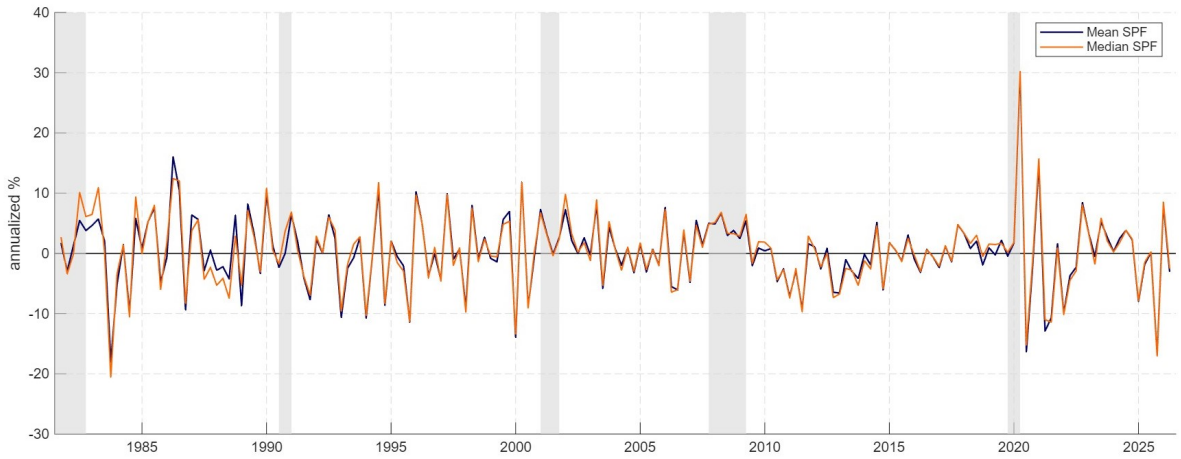


Figure 2: Ramey (2011) forecast error $\Delta g_t - \mathbb{E}^*_{t-1}\Delta g_t$

Notes: 1981Q4–2026Q2, from mean (dark blue) and median (orange) SPF levels. Dates refer to the quarter being forecast. Gray bands are NBER recessions.

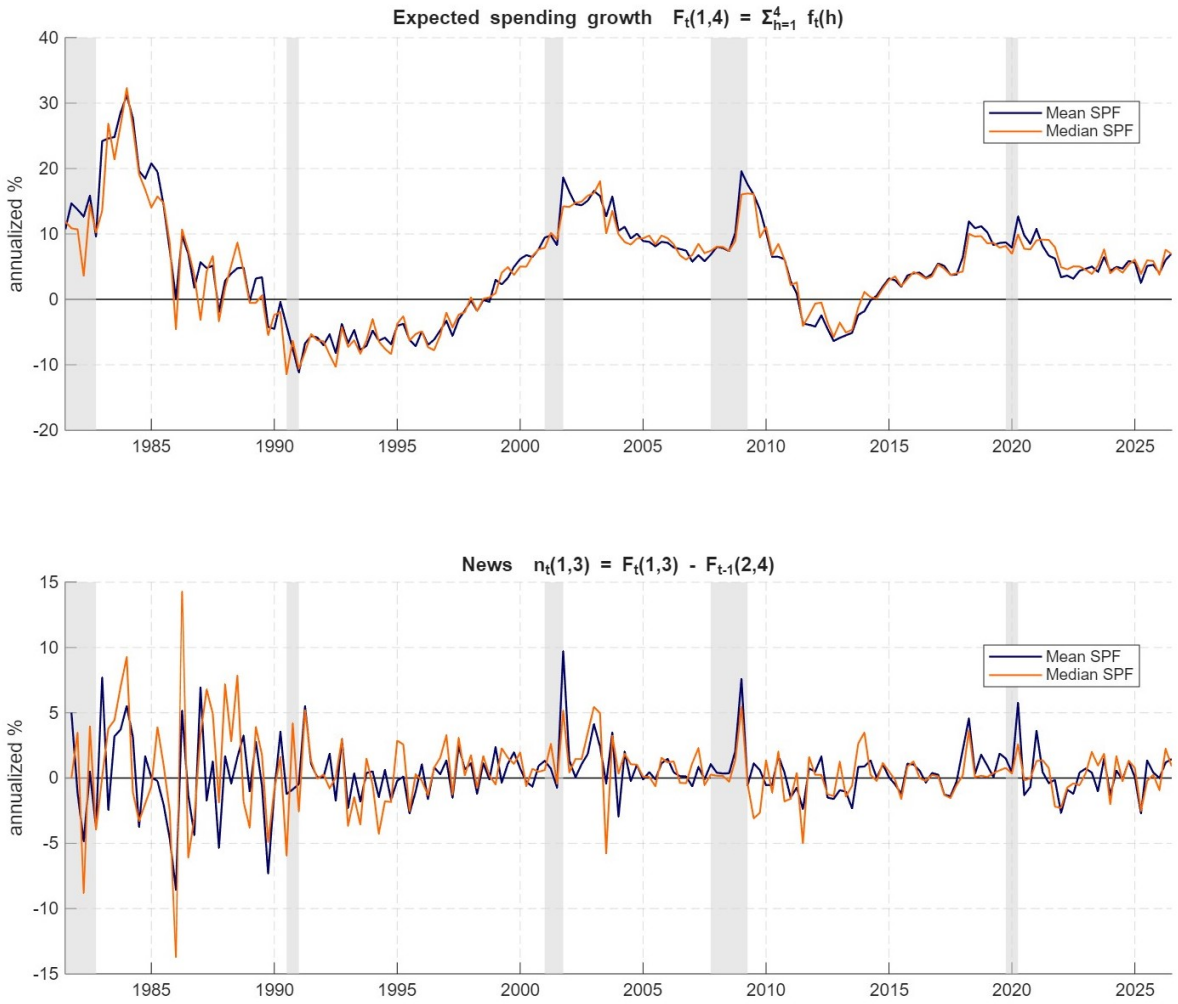


Figure 3: Forni and Gambetti (2016) proxies

Notes: 1981Q3–2026Q3, from mean (dark blue) and median (orange) SPF levels. Top: expected federal spending growth over the next four quarters, $F_t(1,4)$. Bottom: news over the next three quarters, $n_t(1,3)$. Gray bands are NBER recessions.

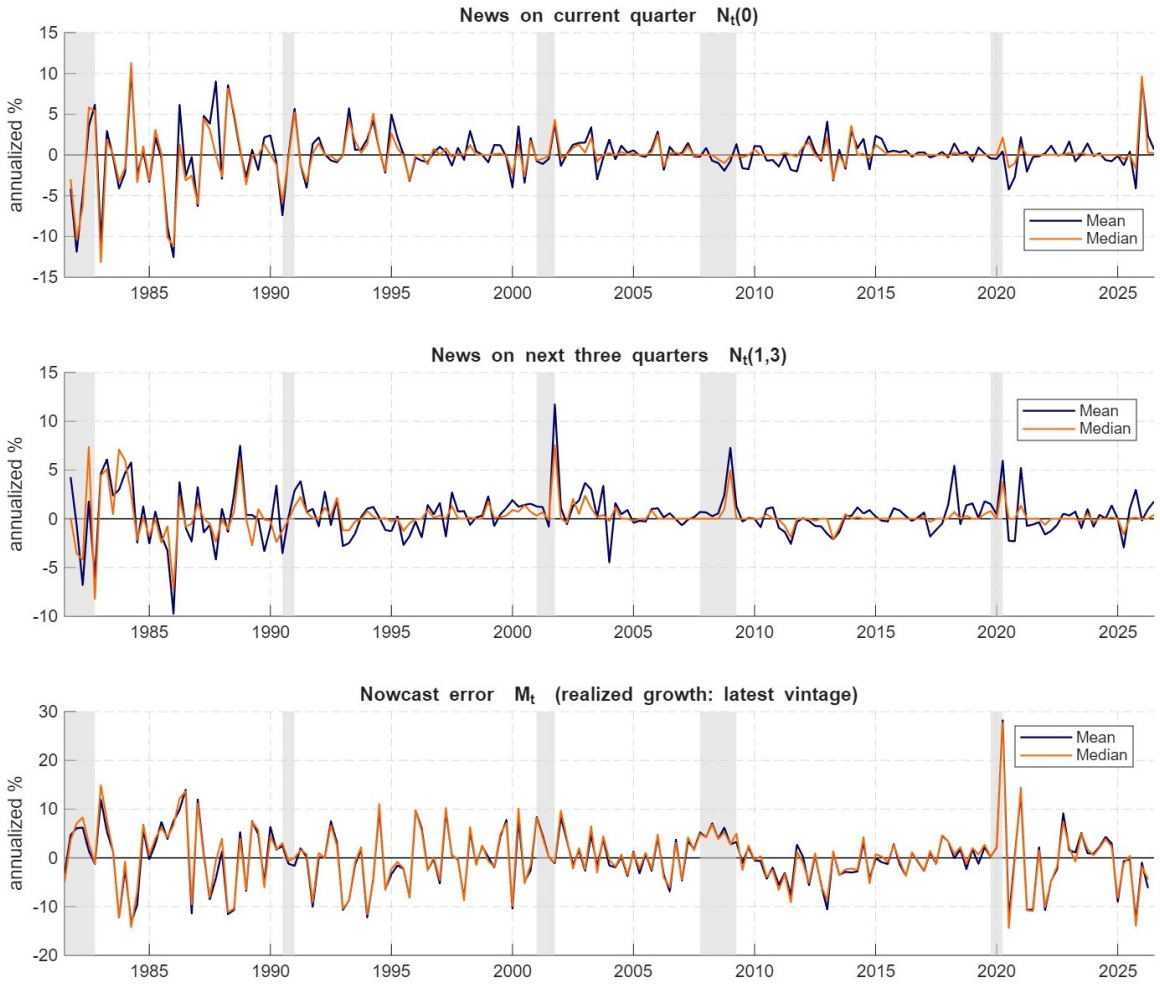


Figure 4: Ricco (2015) proxies

Notes: Ricco (2015) proxies from individual SPF forecasts, aggregated by the mean (dark blue) and the median (orange): news on the current quarter $\mathcal{N}_t(0)$ and news on the next three quarters $\mathcal{N}_t(1,3)$, 1981Q4–2026Q3, and nowcast error $\mathcal{M}_t$, 1981Q3–2026Q2. Revisions use forecasters responding in consecutive surveys. Gray bands are NBER recessions.